

MODEL AND DIE MATERIALS

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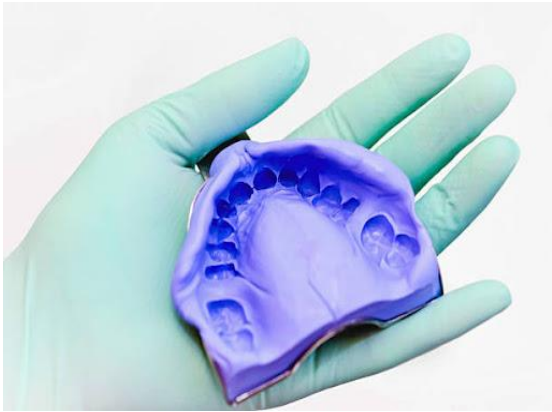
Model & Die Materials

- 1) Gypsum products
- 2) Epoxy products



Definitions

- ▶ **Impression** : A negative replica of oral structures.
- ▶ **A Model or cast** : A positive replica of oral tissues, prepared from an impression .
- ▶ **A Die**: A positive replica of a single tooth ,prepared from an impression.



Requirements of model and die materials:

- 1- It Should have **high strength** to resist breakage during use.
- 2- It Should have **high hardness** to resist scratching during use.
- 3- It should have **good working time**.
4. It should **reproduce fine details** of the impression.
5. It should have **no dimensional changes** during setting and storage.
- 6- It should have **color matching** with impression material.
- 7- **Ease of use**.
- 8- **Cheap**



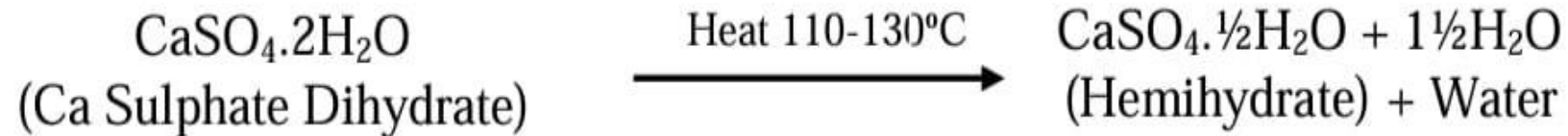
Gypsum products

It is found naturally in the form of Calcium Sulphate dihydrate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$)

- All the forms have the same chemical composition .They differ in the method of manufacturing So they differ in properties and uses.

On heating (Calcination reaction)

Calcium Sulphate dihydrate is converted to Calcium Sulphate hemihydrate.



Gypsum Products

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graph TD; GP[Gypsum Products] --> TI[Type I Impression Plaster]; GP --> TII[Type II Model Plaster]; GP --> TIII[Type III Dental Stone]; GP --> TIV[Type IV High Strength Low Expansion Dental Stone]; GP --> TV[Type V High Strength High Expansion Dental Stone];
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Type I
Impression
Plaster

Type II
Model Plaster

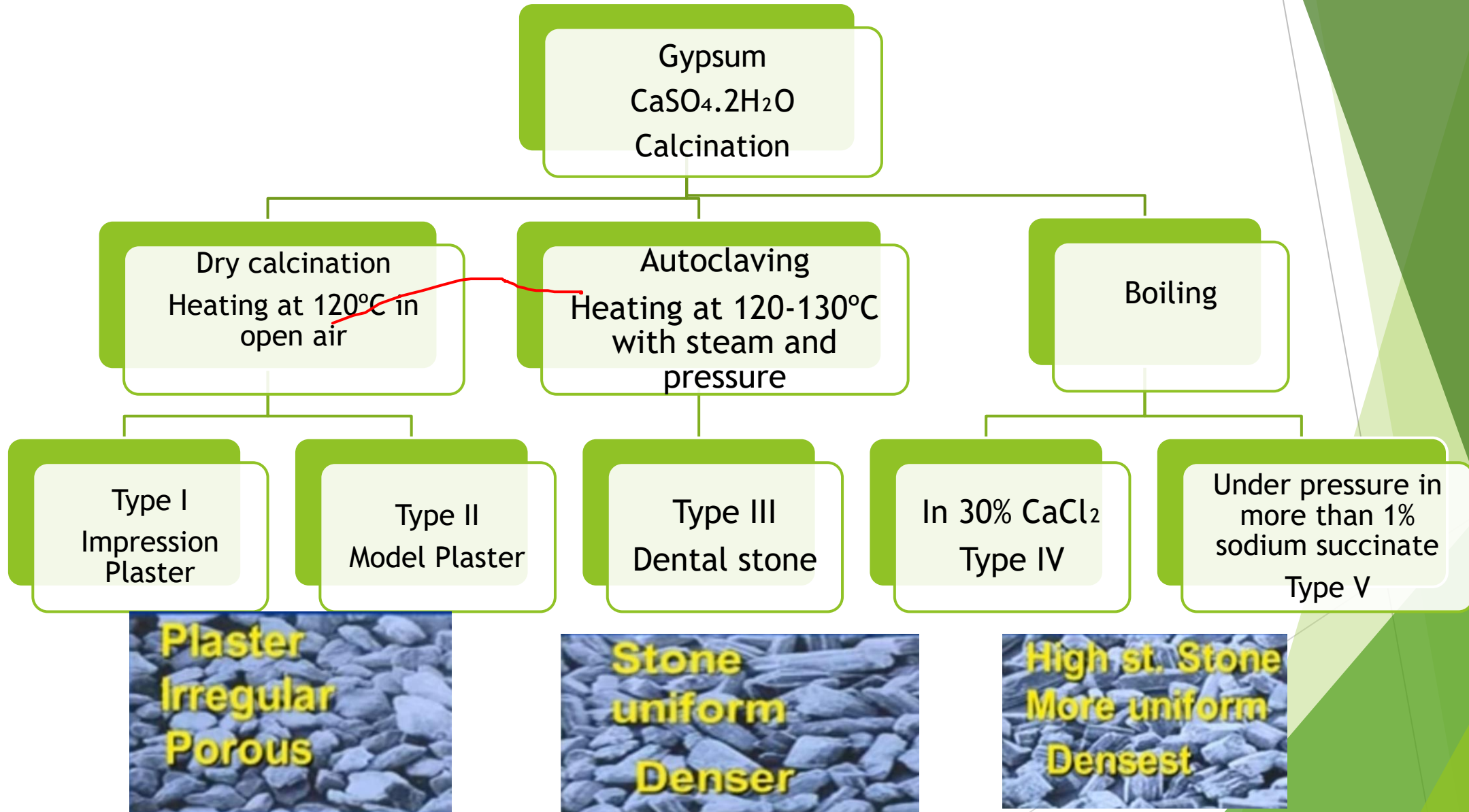
Type III
Dental Stone

Type IV
High Strength
Low Expansion
Dental Stone

Type V
High Strength
High
Expansion
Dental Stone



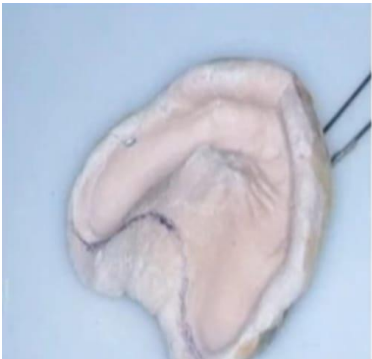
Manufacturing of different types





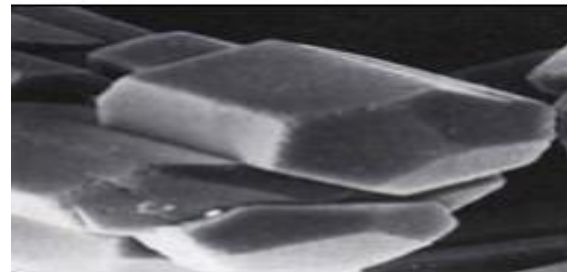
1-Dental plaster:

- ▶ **Manufactured by:** Heating the gypsum in open air at 120 °C
- ▶ **The powder particles obtained are:** Irregular in shape, large and porous.
- ▶ **Uses:** Type I: Impression material.
Type II: primary cast.



2-Dental stone:

- ▶ ***Manufactured by:*** Heating the gypsum at 120 - 130 °C in an autoclave (under steam pressure).
- ▶ ***The powder particles obtained are:*** Smaller, more regular in shape "prismatic in shape", less porous and are denser than that of Type II.
- ▶ ***Uses:*** secondary cast e.g. casts used in fabrication of full denture.



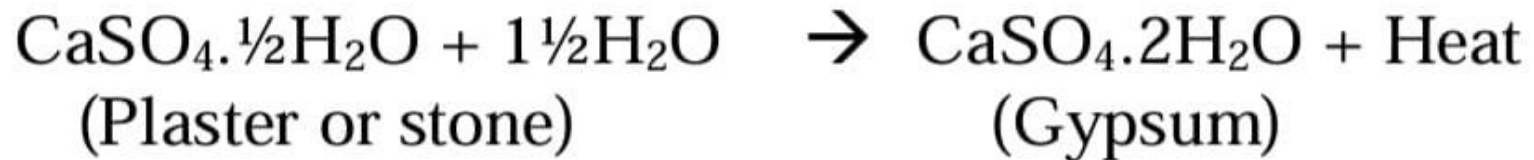
3-High strength stones:

- ▶ **Manufactured by:** Boiling gypsum in 30% calcium chloride (CaCl_2) (Type IV) or under pressure in more than 1% sodium succinate (Type V).
- ▶ **The powder particles obtained are:** The smallest, the most regular, the least porous and are highly dense.
- ▶ **Uses :** *Type IV:* harder and stronger models and single dies
Type V: Special Dies.

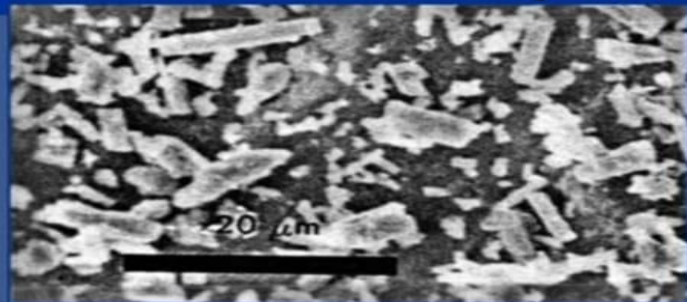


Gypsum products

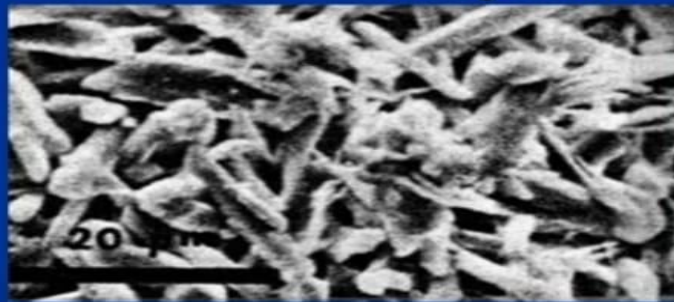
- ▶ The **calcium sulphate hemihydrate** is supplied to dental professional to be mixed with water.
- ▶ The setting reaction is highly **exothermic** (the amount of heat evolved is equal to the heat used for calcination).
- ▶ It obtains the water lost during manufacturing.



Calcium Sulfate **Hemihydrate**



Calcium Sulfate **Dihydrate**



Gypsum products

- ▶ The amount of water of crystallization (essential water)(required for the chemical reaction) for different products is the same (each 100gm of $\text{CaSO}_4 \frac{1}{2} \text{H}_2\text{O}$ requires 18.6 g of water).
- ▶ Practically ,excess water is needed to form homogenous workable mix . This excess water differs among the different gypsum products.
- ▶ N.B. The excess water will evaporate in about 7 days leaving porous material.



Gypsum products

The amount of water required for 100g :

- ▶ Model plaster=55g
- ▶ Dental stone=30g
- ▶ High strength stone=22g



Setting Theory

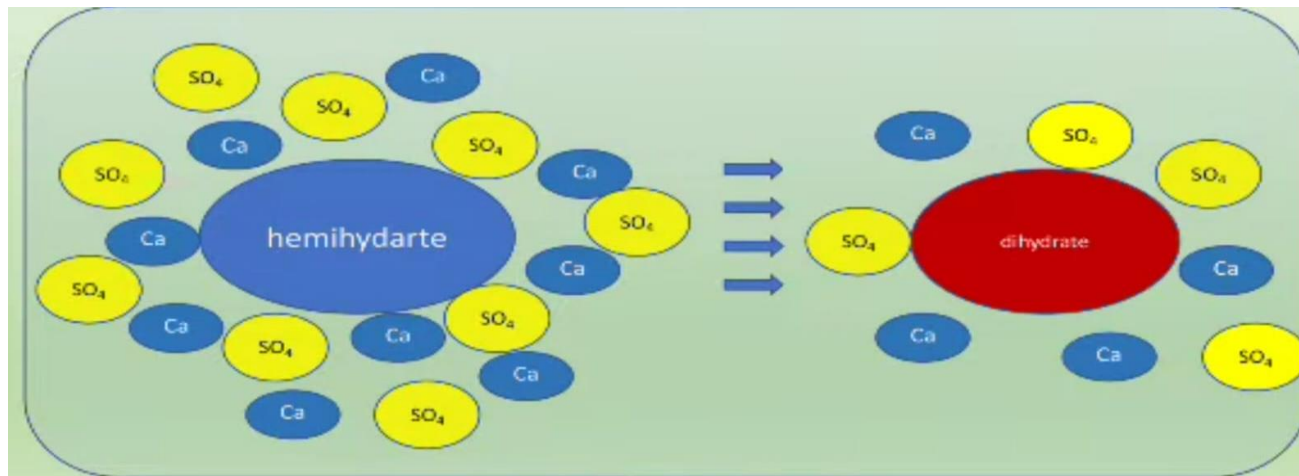
► Crystallization Theory:

It refers setting of Gypsum due to difference in Solubility Rate between $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ and $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

Part of $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ dissolve in water  Ca^{++} and SO_4^{-}

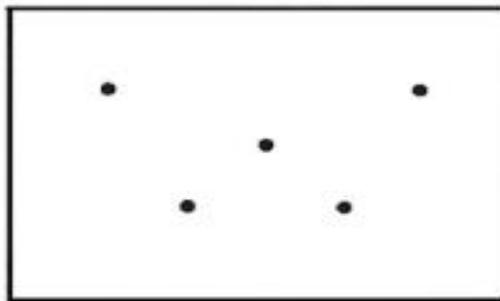
Ca^{++} and SO_4^{-} reacts to form $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

The solubility of $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O} > \text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

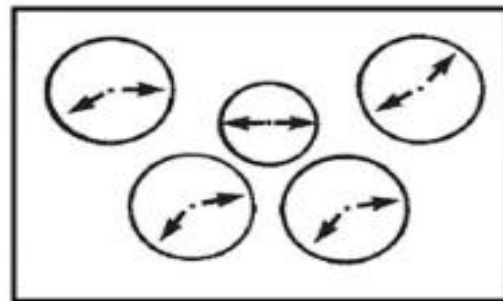


Setting Theory

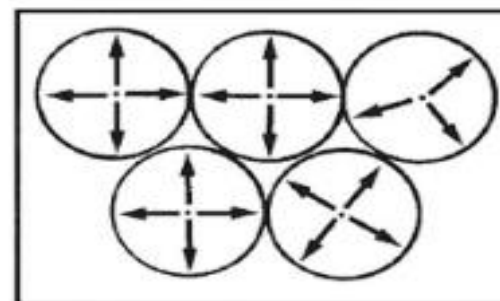
- ▶ The amount of $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ decreased
- ▶ The amount of $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ increased
- ▶ $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ form supersaturated solution precipitated



(a) the nuclei from which crystals grow



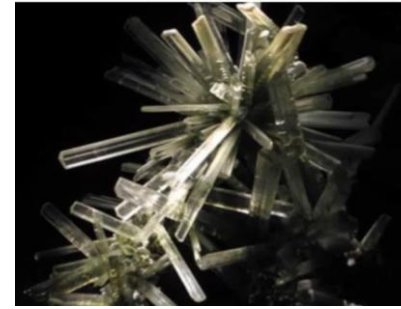
(b) spherulitic growth



(c) the outward thrust as spherulites make contact.

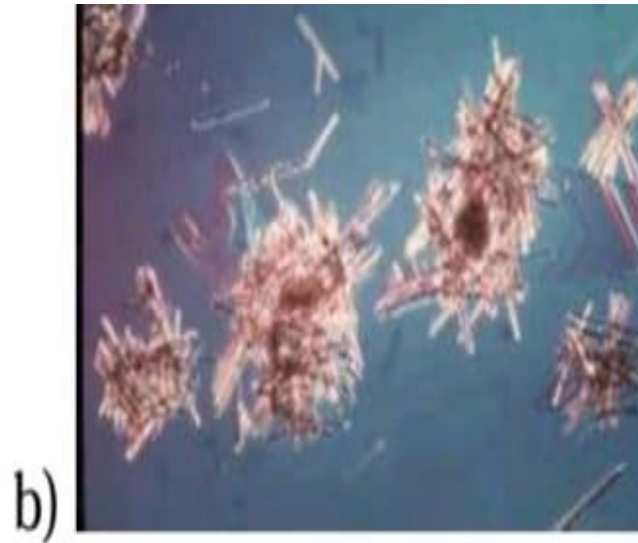


Setting Theory



CaSO₄.2H₂O precipitation starts as nuclei of crystallization

➔ grows to form crystals



Properties

1- Setting time:

- It is the time needed for completion of the setting reaction it is divided into:

Time	From	Until	Minutes (Average)
Mixing time	Add powder to water	Obtain homogenous workable mix	1 minute
Working time		Having even consistency (workable mix)	3 minutes
Initial setting		Material not flow . It can be carved	12 minutes
Ready to use		Can separate the impression	60 minutes
Final setting		Complete the reaction	7 days



Properties

1. Setting time:

Measuring setting time :



a) Loss of surface gloss:

- ▶ The mix starts to lose its gloss. It indicates initial setting. The impression shouldn't be removed at this stage. (Estimate the proper working time).

b) Temperature rise :

- ▶ There is an increase in the temperature as the reaction is exothermic (not accurate method).



Properties

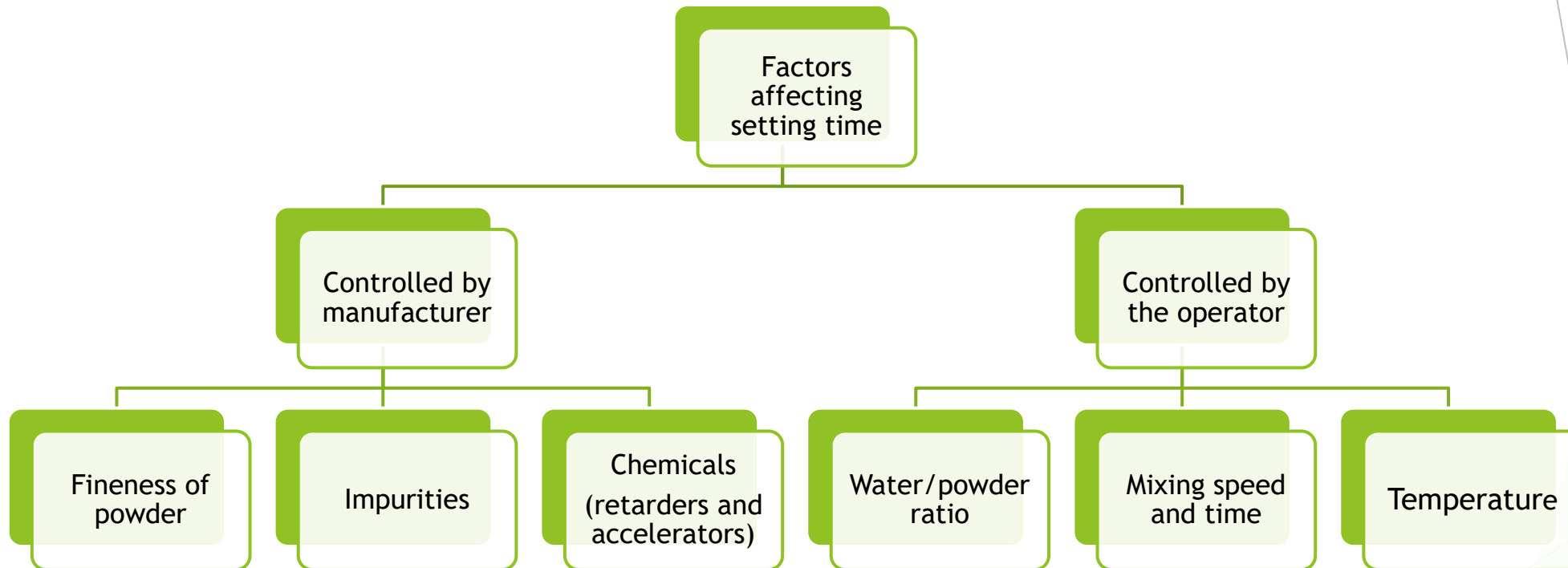
1.Setting Time:

C) penetration test :

- ▶ It is the accurate method to measure the setting time.
- ▶ As the material set it will resist the needle.
- ▶ Vicate test (measure the initial setting time)
- ▶ Gillmore test(measure the initial and final setting).



Properties



Properties

1. Setting Time:

- ▶ Factors controlled by the manufacturer

a) Fineness of the powder:

- ▶ Finer powder(smaller) ➡ accelerate the reaction (decrease setting time).

The faster the rate of dissolution of the hemihydrate. The more the number of nuclei of crystallization



Properties

b) Impurities:

- ▶ Small amount of calcium sulphate dihydrate (Terra alba) 0.5-1%
- ▶ Act as nucleating agent → increasing number of nuclei of crystallization → decrease setting time → accelerate the reaction.
- ▶ N.B. Gypsum products powder should be protected from moisture contamination **Why??** Moisture change some hemihydrate into dihydrate increasing Nuclei of crystallization so decreasing setting time. But after while more & more dihydrate causes reverse effect and increase setting time.



Properties

1. Setting Time:

Factors controlled by the manufacturer

c) Chemicals:

► Retarder (increase setting time):

i) Makes Coating on hemihydrate so reducing its solubility OR coating on dehydrate crystals (poisoning of the nuclei).

e.g. Organic materials as Glue, Gelatin, some Gums, dry blood or Saliva.

ii) Formation of a layer of calcium salt which is less soluble than sulphate.

e.g. Borax is the most commonly used retarder.



Properties

1. Setting Time:

Factors controlled by the manufacturer

c) Chemicals:

► Accelerator (decrease setting time) :

i) Increasing rate of solubility of hemihydrate.

ii) Addition of more nuclei of crystallization(Most soluble Inorganic salts ,in small amounts act as accelerator, in large amount act as retarder as they deposit on surface of crystals).

e.g. potassium sulphate up to 2% is the most commonly used accelerator



Properties

1. Setting Time:

Factors controlled by the operator

a) Water / Powder ratio:

- ▶ High W/P = watery mix  retardation
- ▶ Low W/P = thick mix  acceleration



Properties

1. Setting Time:

Factors controlled by the operator

b) Mixing speed and time(within limit):

- ▶ Increasing in spatulation (speed or time or both) → Accelerate the reaction (breaking up some dihydrate crystals so increasing the number of nuclei of crystallization).



Properties

1. Setting Time:

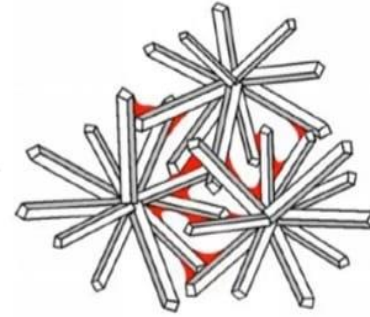
Factors controlled by the operator

c) Water temperature:

- ▶ 20:50 °C → accelerate the reaction
- ▶ Above 50 °C → retard the reaction
- ▶ 100 °C → no reaction (no setting occurs as solubility of hemihydrate = solubility of Dihydrate)



Properties



- ▶ Theoretically, the volume of the final dihydrate should be about 7% less than the total volume of hemihydrate and water entering the reaction.
- ▶ Actually, practically expansion is actually observed during setting.
- ▶ This expansion is due to outward thrusting action of the growing crystals(the crystals pushing each other apart during their growth).



Properties:

2) Dimensional changes:

a) setting expansion:

The setting expansion of different types of Gypsum:

- ▶ Plaster=0.3%
- ▶ Dental stone=0.08%
- ▶ Improved stone=0.06%



Properties:

2) Dimensional changes:

a) Setting expansion:

Factors controlling setting expansion:

i) W/P ratio:

low W/P ratio increase setting expansion. As more nuclei of crystallization will be available per unit volume i.e. less space between the nuclei. Consequently, more growth interaction of dihydrate crystals will take place with more outward thrust and hence more expansion.

ii. Mixing (Spatulation) speed and time: **An increase in spatulation** (speed and/or time or both) within limits **increase setting expansion.** due to the formation of more nuclei of crystallization.

iii. Chemicals: Some Chemicals decrease the setting expansion by regulating the shape of the growing crystals and thus decrease the thrusting action.



Properties:

2) Dimensional changes:

b) Hygroscopic Expansion:

- ▶ It occurs if the gypsum is allowed to set under water during initial setting.
- ▶ It may reach double the setting expansion.
- ▶ **Why???** This is because the additional water provides more room for crystal growth; therefore, the crystals can grow more freely resulting in a greater degree of expansion.



Properties:

2) Dimensional changes:

b) Hygroscopic Expansion:

Factors controlling hygroscopic expansion:

- i. **Time of immersion in water:** Increasing the time of immersion in water during the initial stage of setting will increase the hygroscopic expansion.
- ii. **Amount of water added:** The more water added at the initial setting time, the higher the hygroscopic expansion.
- iii. **W/P ratio of the original mix:** The lower the W/P ratio of the original mix, the higher the hygroscopic expansion.



Properties:

3) Strength:

- ▶ Gypsum products **are brittle materials.**
- ▶ There are two types of strength according to presence of excess water
 - a) Wet strength
 - b) dry strength



Properties:

3) Strength:

a) Wet strength (1-hour strength)(green strength):

- ▶ It is the strength with excess water.

b) Dry strength (7-days strength):

- ▶ It is the strength after evaporation of water.
- ▶ It is double wet strength.



Properties:

3) Strength:

► Factors affecting strength:

1- The higher water/powder ratio the weaker will be the Gypsum.

2- Increasing the mixing time and rate within limits will provide more nuclei of crystallization thus strength will be increased.

3-The type of gypsum products: Improved and dental stone are stronger than model plaster.



Properties

3) Strength:

► Factors affecting strength:

4-Time: Dry strength is higher than the wet strength.

5- Chemicals "accelerators + retarders": They decrease the strength by reducing the inter-crystalline cohesion

► However, over-mixing will result in decreasing the strength of the final product , **why???** This is because the formed gypsum crystals will be broken up and less crystalline interlocking will occur.



Properties:

4) Surface Hardness:

- ▶ Gypsum products have low surface hardness and liable to scratching
- ▶ Improved and dental stone are harder than model plaster.
- ▶ Incorporation of small amounts of resin "polymer", will increase hardness and improve abrasion resistance.



Properties:

5) Reproduction of details:

- ▶ Gypsum dies do not reproduce surface details very well if:
 - Air bubbles are formed at the interface of the impression and gypsum cast.
- ▶ The use of vibrator during pouring of the cast reduces the presence of air bubbles.



Comparison of different types of gypsum products

Difference	Plaster		Dental stone	High strength stone	
Method of production	Open air		Autoclave	<u>Type IV</u> (Low expansion) Boiling in 30% CaCl ₂	<u>Type V</u> (High expansion) Pressure + More than 1% sodium succinate
Particle size and shape	Irregular, porous		Regular, less porous	Most regular and least porosity	
Hardness	15 R.H.N.		60 R.H.N.	80 R.H.N.	
Porosity %	45 %		15 %	10 %	
Uses (Examples)	<u>Type I</u> Impression material.	<u>Type II</u> - Flasking - Primary cast	- Secondary cast. - Binder in investment material	<u>Type IV</u> Secondary cast and dies for crown & bridge work.	<u>Type V</u> Special dies, when higher die expansion is needed.



Thank You

